CSET485- AI and Society: Assignment 1

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Assigned Dataset: ***German Credit***

Data Random Seed (Train/Test Split): ***195***

Data Preprocessing

I used the German Credit dataset, which contains 1,000 records and 20 input features. First, I checked for missing values, and there were none.

I used One-Hot Encoding for categorical data and StandardScaler for numerical features. I checked for outliers using the IQR method but did not remove them, as they could be actual values.

The data was split into 70% training and 30% testing using random seed 195. After preprocessing, I got 48 features, which were used to train and test both models.

```python
import pandas as pd
import numpy as np

# Column names from the German Credit dataset
columns = [
    "A1", "A2", "A3", "A4", "A5",
    "A6", "A7", "A8", "A9", "A10",
    "A11", "A12", "A13", "A14", "A15",
    "A16", "A17", "A18", "A19", "A20",
    "class"
]

# Read the whitespace-separated data
df = pd.read_csv(
    "german.data",
    sep=r"\s+",
    header=None,
    names=columns
)

# Original class: 1 = good credit, 2 = bad credit
# Assignment target: 1 = approve, 0 = reject
df["approval"] = (df["class"] == 1).astype(int)
df = df.drop(columns=["class"])

print("Dataset shape:", df.shape)
print("\nFirst five rows:")
display(df.head())

print("\nApproval distribution:")
print(df["approval"].value_counts())

print("\nMissing values:")
print(df.isnull().sum().sum())
```

```
Dataset shape: (1000, 21)

First five rows:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	...	A12	A13	A14	A15	A16	A17	A18	A19	A20	approval
0	A11	6	A34	A43	1169	A65	A75	4	A93	A101	...	A121	67	A143	A152	2	A173	1	A192	A201	1
1	A12	48	A32	A43	5951	A61	A73	2	A92	A101	...	A121	22	A143	A152	1	A173	1	A191	A201	0
2	A14	12	A34	A46	2096	A61	A74	2	A93	A101	...	A121	49	A143	A152	1	A172	2	A191	A201	1
3	A11	42	A32	A42	7882	A61	A74	2	A93	A103	...	A122	45	A143	A153	1	A173	2	A191	A201	1
4	A11	24	A33	A40	4870	A61	A73	3	A93	A101	...	A124	53	A143	A153	2	A173	2	A191	A201	0

5 rows × 21 columns

```
Approval distribution:
approval
1    700
0    300
Name: count, dtype: int64

Missing values:
0
```

2. Part 1: Build

I trained Linear Regression and Logistic Regression on the dataset using 70% data for training and 30% for testing.

Linear Regression used a threshold of 0.7359, while Logistic Regression used 0.5. Both models gave different results for 82 applicants (27.33%).

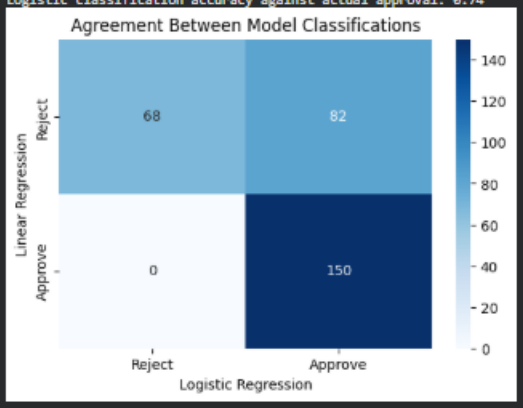
Linear Regression achieved 65.33% accuracy, while Logistic Regression achieved 74%. So, Logistic Regression performed better on this dataset.

```
Linear Regression median threshold: 0.7359
Logistic Regression threshold: 0.5

Confusion Matrix:
Rows = Linear Regression; Columns = Logistic Regression
[[ 68  82]
 [  0 150]]

Mismatched cases: 82
Mismatch percentage: 27.33 %

Linear classification accuracy against actual approval: 0.6533
Logistic classification accuracy against actual approval: 0.74
```



3. Part 2: The Trap

2.1 Simpson's Paradox Hunt

I divided the applicants into two age groups: under 30 and 30 or above. I compared the effect of credit duration on approval for both groups and the full dataset.

The coefficient was -0.03162 for the full dataset, -0.03173 for applicants under 30, and -0.03254 for applicants aged 30 or above. All three values were negative, so I did not find a reversal in the trend. The results were very similar across the groups.

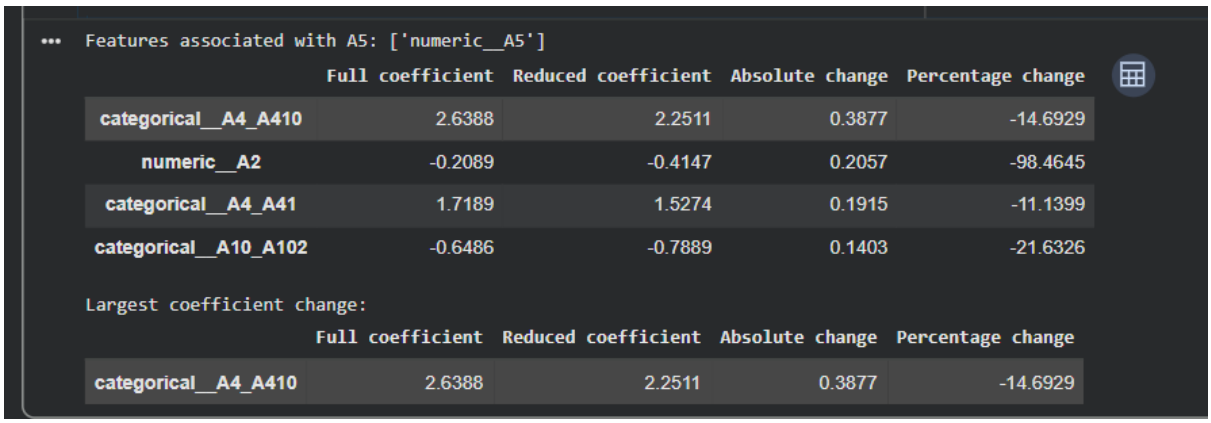
```
--- Full training dataset ---
Number of applicants: 700
Duration coefficient (A2): -0.03162
95% confidence interval: (-0.04502, -0.01823)
P-value: 0.0

--- Applicants under 30 ---
Number of applicants: 268
Duration coefficient (A2): -0.03173
95% confidence interval: (-0.0537, -0.00976)
P-value: 0.00464

--- Applicants aged 30 and above ---
Number of applicants: 432
Duration coefficient (A2): -0.03254
95% confidence interval: (-0.04959, -0.01549)
P-value: 0.00018
```

2.2 Omitted-Variable Bias

I removed the credit amount variable (A5) and compared the coefficients before and after. The coefficient of A2 changed from -0.2089 to -0.4147, which is a 98.46% change in magnitude. The A4_A410 coefficient had the largest absolute change of 0.3877. This shows that removing a variable can affect other model results.



Features associated with A5: ['numeric__A5']

	Full coefficient	Reduced coefficient	Absolute change	Percentage change
categorical__A4_A410	2.6388	2.2511	0.3877	-14.6929
numeric__A2	-0.2089	-0.4147	0.2057	-98.4645
categorical__A4_A41	1.7189	1.5274	0.1915	-11.1399
categorical__A10_A102	-0.6486	-0.7889	0.1403	-21.6326

Largest coefficient change:

	Full coefficient	Reduced coefficient	Absolute change	Percentage change
categorical__A4_A410	2.6388	2.2511	0.3877	-14.6929

2.3 Adversarial Subset Construction

The model made 33 high-confidence wrong predictions, which is 11% of the test data. Since there were fewer than 50 such cases, I used all 33 for analysis. Their average age was 35.36 years, which was close to the test set average of 35.60 years. The A93 category made up 51.52% of these cases, compared to 58% of the full test set.

```
High-confidence wrong predictions: 33
Percentage of full test set: 11.0 %

Rows selected for adversarial analysis: 33

Age distribution: adversarial subset
count    33.000000
mean     35.363636
std      12.633829
min      21.000000
25%      25.000000
50%      31.000000
75%      42.000000
max      66.000000
Name: A13, dtype: float64

Age distribution: full test set
count    300.000000
mean      35.603333
std       10.986126
min       20.000000
25%       27.000000
50%       33.000000
75%       42.000000
max       75.000000
Name: A13, dtype: float64

Gender breakdown: adversarial subset
A9
A93    51.52
A92    27.27
A94    18.18
A91     3.03
Name: proportion, dtype: float64

Gender breakdown: full test set
A9
A93    58.00
A92    28.67
A94     9.33
A91     4.00
Name: proportion, dtype: float64

First five adversarial cases:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	...	A13	A14	A15	A16	A17	A18	A19	A20	actual	predicted_probability
80	A14	24	A32	A43	5943	A65	A72	1	A92	A101	...	44	A143	A152	2	A173	1	A192	A201	0	0.922929
17	A11	30	A30	A49	8072	A65	A72	2	A93	A101	...	25	A141	A152	3	A173	1	A191	A201	1	0.095165
861	A14	18	A33	A49	2169	A61	A73	4	A94	A101	...	28	A143	A152	1	A173	1	A192	A201	0	0.806490
137	A12	12	A32	A43	766	A63	A73	4	A93	A101	...	66	A143	A152	1	A172	1	A191	A201	0	0.930901
417	A11	18	A33	A46	8471	A65	A73	1	A92	A101	...	23	A143	A151	2	A173	1	A192	A201	1	0.248657

5 rows × 22 columns

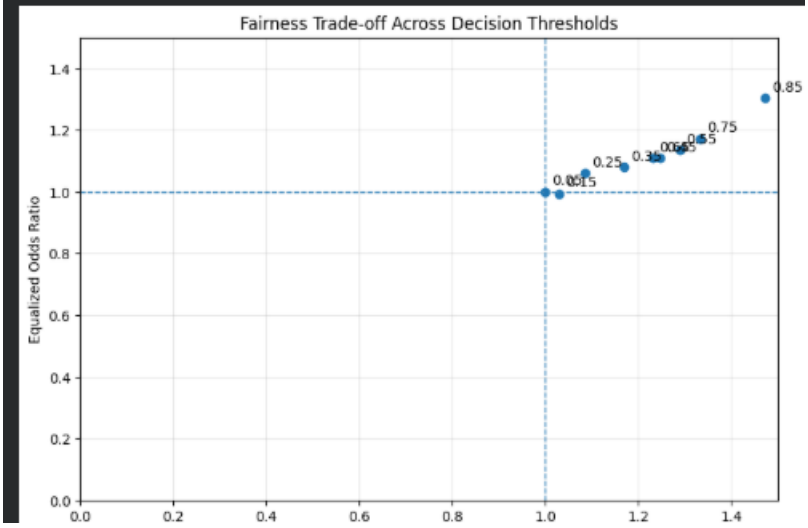
2.4 Fairness Trade-Off Analysis

I compared male and female groups using three fairness metrics at different thresholds. At threshold 0.15, the DP ratio was 1.0292, EO was 0.9935, and PP was 1.0413. At threshold 0.85, DP increased to 1.4735 and EO to 1.3039, while PP was 0.9545. This shows that changing the threshold affects the fairness results, and no single threshold is best for every metric.

```
Group counts:
group
Male      214
Female     86
Name: count, dtype: int64

Fairness metrics at each threshold:
```

	Threshold	Demographic Parity	Equalized Odds	Predictive Parity
0	0.05	1.0000	1.0000	1.0787
1	0.15	1.0292	0.9935	1.0413
2	0.25	1.0881	1.0603	1.0531
3	0.35	1.1696	1.0804	0.9964
4	0.45	1.2458	1.1097	0.9609
5	0.55	1.2890	1.1346	0.9495
6	0.65	1.2307	1.1083	0.9714
7	0.75	1.3325	1.1724	0.9491
8	0.85	1.4735	1.3039	0.9545
9	0.95	1.6521	NaN	0.9730



Part 3: Descriptive Analysis Questions

Q1. Difference between Linear Regression and Logistic Regression

The two models were trained using 700 records and tested using 300. Linear Regression had a threshold of 0.7359, while Logistic Regression had a threshold of 0.5.

Both models gave the same result for 218 applicants, and for 82 applicants (27.33%), the results were different. Linear Regression got 65.33% accuracy, while Logistic Regression got 74% accuracy.

The difference was that the two models work differently and use different thresholds. Logistic Regression performed better on this dataset. This helped me understand how model selection may affect credit approval decisions.

Q2. Simpson's Paradox

I divided the applicants into two age groups: under 30 and 30 or above. Then I compared the effect of credit duration on approval in both groups and the full dataset.

The coefficient was -0.03162 for the full dataset, -0.03173 for applicants under 30, and -0.03254 for applicants aged 30 or above. All values were negative and very similar, so I did not find any reversal in the trend.

This shows that the relationship between credit duration and approval remained almost the same across both age groups in my analysis.

```
...
    --- Full dataset ---
    Duration coefficient, duration only: -0.03165
    Duration coefficient, adjusted for age: -0.03162
    P-value, duration only: 0.0
    P-value, adjusted model: 0.0

    --- Under 30 ---
    Duration coefficient, duration only: -0.03104
    Duration coefficient, adjusted for age: -0.03173
    P-value, duration only: 0.00524
    P-value, adjusted model: 0.00464

    --- Age 30 and above ---
    Duration coefficient, duration only: -0.03304
    Duration coefficient, adjusted for age: -0.03254
    P-value, duration only: 0.00014
    P-value, adjusted model: 0.00018
```

Q3. Omitted Variable Bias

I removed the credit amount variable (A5) and compared the model coefficients before and after removing it.

The coefficient of A2 changed from -0.2089 to -0.4147, showing a 98.46% change in magnitude. The A4_A410 coefficient had the largest absolute change of 0.3877.

This shows that removing a variable can affect other coefficients in the model. It helped me understand why choosing relevant variables is important for getting reliable results.

```
...  Features associated with A5: ['numeric__A5']
```

	Full coefficient	Reduced coefficient	Absolute change	Percentage change
categorical__A4_A410	2.6388	2.2511	0.3877	-14.6929
numeric__A2	-0.2089	-0.4147	0.2057	-98.4645
categorical__A4_A41	1.7189	1.5274	0.1915	-11.1399
categorical__A10_A102	-0.6486	-0.7889	0.1403	-21.6326

```
Largest coefficient change:
```

	Full coefficient	Reduced coefficient	Absolute change	Percentage change
categorical__A4_A410	2.6388	2.2511	0.3877	-14.6929

Q4. Adversarial Subset Analysis

The model made 33 high-confidence wrong predictions, which was 11% of the test data. Since there were fewer than 50 cases, I used all 33 for the analysis.

The average age of these applicants was 35.36 years, which was close to the test dataset average of 35.60 years. The A93 category made up 51.52% of these cases, compared to 58% in the full test dataset.

This helped me understand that even when a model is confident, it can still make incorrect predictions.

```
High-confidence wrong predictions: 33
Percentage of full test set: 11.0 %

Rows selected for adversarial analysis: 33

Age distribution: adversarial subset
count    33.000000
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std      12.633829
min      21.000000
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Gender breakdown: adversarial subset
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A91     3.03
Name: proportion, dtype: float64

Gender breakdown: full test set
A9
A93    58.00
A92    28.67
A94     9.33
A91     4.00
Name: proportion, dtype: float64

First five adversarial cases:
```

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	...	A13	A14	A15	A16	A17	A18	A19	A20	actual	predicted_probability
80	A14	24	A32	A43	5943	A65	A72	1	A92	A101	...	44	A143	A152	2	A173	1	A192	A201	0	0.922929
17	A11	30	A30	A49	8072	A65	A72	2	A93	A101	...	25	A141	A152	3	A173	1	A191	A201	1	0.095165
861	A14	18	A33	A49	2169	A61	A73	4	A94	A101	...	28	A143	A152	1	A173	1	A192	A201	0	0.806490
137	A12	12	A32	A43	766	A63	A73	4	A93	A101	...	66	A143	A152	1	A172	1	A191	A201	0	0.930901
417	A11	18	A33	A46	8471	A65	A73	1	A92	A101	...	23	A143	A151	2	A173	1	A192	A201	1	0.248657

5 rows × 22 columns

Q5. Fairness Metrics and Thresholds

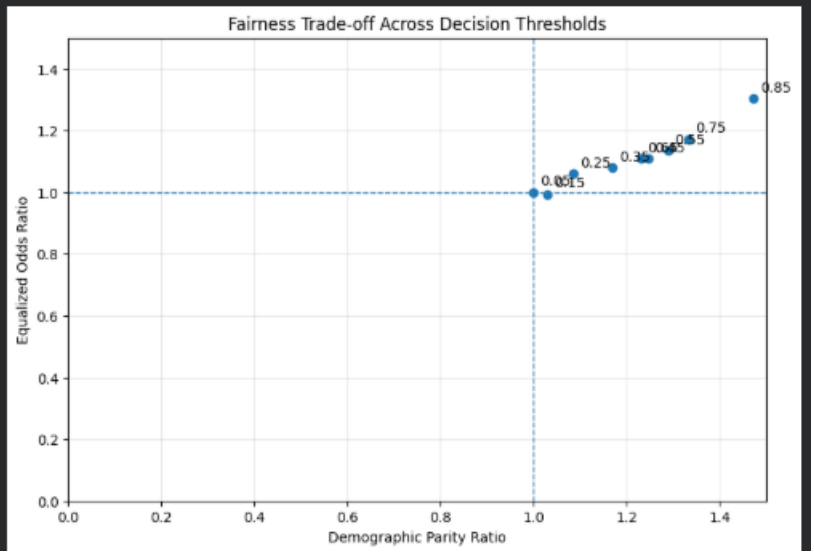
I compared male and female groups using three fairness metrics at different thresholds. At a threshold of 0.15, the DP ratio was 1.0292, EO was 0.9935, and PP was 1.0413.

At a threshold of 0.85, the DP ratio increased to 1.4735 and EO to 1.3039, while PP was 0.9545. This shows that changing the threshold affects the fairness results. No single threshold gave the best result for all three metrics.

```
group
Male      214
Female     86
Name: count, dtype: int64

Fairness metrics at each threshold:
```

	Threshold	Demographic Parity	Equalized Odds	Predictive Parity
0	0.05	1.0000	1.0000	1.0787
1	0.15	1.0292	0.9935	1.0413
2	0.25	1.0861	1.0603	1.0531
3	0.35	1.1696	1.0804	0.9964
4	0.45	1.2458	1.1097	0.9609
5	0.55	1.2890	1.1346	0.9495
6	0.65	1.2307	1.1083	0.9714
7	0.75	1.3325	1.1724	0.9491
8	0.85	1.4735	1.3039	0.9545
9	0.95	1.6521	NaN	0.9730



Q6. Fairness Trade-off in Real Life

In my analysis, I found that changing the decision threshold affected the fairness metrics. At a threshold of 0.15, the DP ratio was 1.0292 and EO was 0.9935. At 0.85, DP increased to 1.4735 and EO increased to 1.3039. This shows that changing the threshold can affect how the two groups are treated.

For example, imagine two applicants from different gender groups who have similar financial situations. At a lower threshold, both may get credit approval, but at a higher threshold, one applicant may be rejected while the other is approved because their predicted scores are different. This can create unfair outcomes for people who are equally deserving.

If we focus only on demographic parity, we may improve the approval rates between groups but still create differences in error rates. Similarly, focusing on equalized odds may not give both groups equal approval rates.

This helped me understand that choosing a threshold is not only a technical decision. It can affect real people, so we need to consider different fairness metrics before using the model.

Q7. Choosing a Fairness Threshold

I think a threshold of 0.15 is a reasonable choice because it gives 72% accuracy, a DP ratio of 1.0292, and an EO ratio of 0.9935. These values are close to 1, which is the ideal value for both fairness metrics.

However, this threshold may not be the best choice for everyone. A critic could argue that overall accuracy is still only 72% and other thresholds may perform better on certain measures. I agree that no single threshold can guarantee fairness in every situation.

Therefore, I would consider 0.15 as a possible option, but I would test it further before using it for real credit approval decisions.

Q8. Demographic Parity vs Equalized Odds

I would recommend prioritizing equalized odds because, in credit approval, it is important to reduce differences in prediction errors between groups. At a threshold of 0.15, the EO ratio was 0.9935 and the DP ratio was 1.0292, both close to the ideal value of 1.

At a threshold of 0.85, the EO ratio increased to 1.3039 and the DP ratio to 1.4735, showing a bigger difference from the ideal value.

By prioritizing equalized odds, we may not achieve exactly equal approval rates for both groups. However, I think reducing differences in errors is important because incorrect credit decisions can affect people's lives. The final choice should still consider accuracy and the impact on applicants.

Part 4: Reflection

Deployment Recommendation

After doing this analysis, I would not recommend using this model alone to make final credit approval decisions. Logistic Regression performed better than Linear Regression, with 74% accuracy compared to 65.33%. However, this does not mean that the model will always make the correct decision. It made 33 high-confidence wrong predictions, which were 11% of the test data.

I think the model could be used as a support tool for bank employees, but the final decision should involve human review, especially for applicants who may be wrongly rejected. Before using it in real life, the model should also be tested on more data.

Why One Number Is Not Enough

During this assignment, I understood that accuracy alone cannot tell us whether a model is reliable or fair.

First, both models gave different results for 82 applicants, which was 27.33% of the test data. This means that the choice of model and threshold can change the decision for many people.

Second, the fairness results changed with the threshold. At 0.15, the demographic parity ratio was 1.0292 and the equalized odds ratio was 0.9935. These values were close to 1. At 0.85, they increased to 1.4735 and 1.3039. So, a threshold that looks acceptable for one fairness measure may give worse results for another.

Third, the model made 33 high-confidence wrong predictions. This shows that even when a model is confident, its prediction can still be incorrect. We cannot blindly trust its output.

Finally, removing the credit amount variable changed the coefficients of other variables. The magnitude of the A2 coefficient changed by 98.46%. This showed me that the variables included in a model can affect how we understand its results.

Evidence and Final Thoughts

The results from this assignment showed me that building a model is not just about getting good accuracy. We also need to check how its decisions affect different groups, whether its predictions can be trusted, and whether the results change when we modify the data or model.

I would also want to test the model on new data and check its performance regularly before considering real-world use. Credit decisions can affect people's lives, so mistakes should not be ignored just because the overall accuracy looks good.

Overall, this assignment helped me understand that a model should be judged using multiple measures instead of one number. A model may perform well technically but still have problems when used in real situations. Human review and regular testing are important before relying on it for important decisions.

Colab code link: https://colab.research.google.com/drive/1tjiYYFAwZ3QXAa4DS2cVXu8-osH_J3tI?usp=sharing

GitHub Link: https://github.com/ARPIT-KUMAR-SAHU1/AI-and-Society-Assignment